

- 7 (a) 5 ohms A1
6, 6 A1
- (b) (i) Decay constant is the probability of decay per unit time of a nucleus. B1
- (ii) Half-life is the time taken for half the (number of radioactive nuclei/count rate/activity) present in any given sample of a given isotope to decay at any given time. B2
- (c) Radioactive decay is a random process, in other words we don't know which nuclei will decay next; we only know the probability of decay.

After one half-life, it is not guaranteed that exactly half of the original atoms remain, but that this is just the most likely, and the average outcome.

Marks awarded as follow:

- meaning of random M1
- a reasonable discussion of what that means. A1

(d) (i) Original ratio = $\frac{C_{14,initial}}{C_{12,initial}} = \frac{1}{3.3 \times 10^{10}}$

Over the years, the ratio became $\frac{C_{14,final}}{C_{12,initial}} = \frac{1}{8.6 \times 10^{10}}$

$$\frac{N}{N_0} = \frac{C_{14,final}}{C_{14,initial}} = \frac{C_{14,final}}{C_{12,initial}} \times \frac{C_{12,initial}}{C_{14,initial}} = \frac{1}{8.6 \times 10^{10}} \times \frac{3.3 \times 10^{10}}{1}$$

C1

$$= 0.384$$

A0

Assumption: Ratio of C-12 to C-14 for a fresh sample of wood is constant/ C-12 remains the same B1

(ii) $N = N_0 e^{-\lambda t}$

$$\lambda = \frac{\ln 2}{5700} = 1.2160 \times 10^{-4} \text{ y}^{-1}$$

$$0.384 = e^{-1.2160 \times 10^{-4}(t)}$$

M1

$$t = 2.49 \times 10^{11} \text{ s} = 7870 \text{ years}$$

A1

8 (a) larger energy gap between conduction band and lower orbitals than standard silicon M1